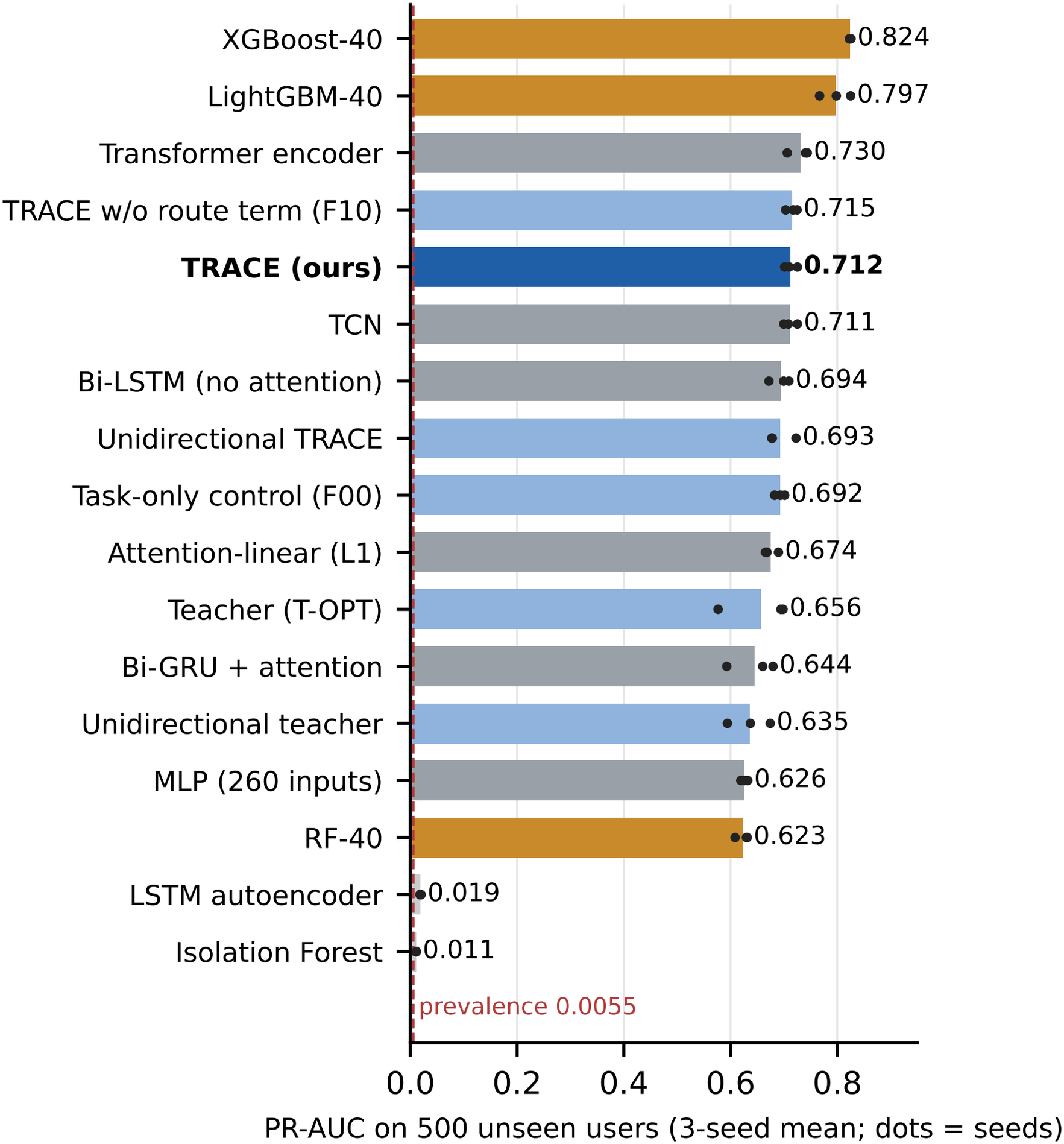
XGBoost-40
LightGBM-40
Transformer encoder
TRACE w/o route term (F10)
TRACE (ours)
TCN
Bi-LSTM (no attention)
Unidirectional TRACE
Task-only control (F00)
Attention-linear (L1)
Teacher (T-OPT)
Bi-GRU + attention
Unidirectional teacher
MLP (260 inputs)
RF-40
LSTM autoencoder
Isolation Forest
0.824
0.797
0.730
0.715
0.712
0.711
0.694
0.693
0.692
0.674
0.656
0.644
0.635
0.626
0.623
0.019
0.011
prevalence 0.0055
0.0
0.2
0.4
0.6
0.8
PR-AUC on 500 unseen users (3-seed mean; dots = seeds)